Heriot-Watt University

Department of Actuarial Mathematics and Statistics

Life Insurance Mathematics B

Tutorial 9

1. Calculate the following on the basis of PMA92/PFA92 mortality and interest of 4% (in each case the first age is of a male life and the second age is of a female life):

   (a) $\ddot{a}_{70|70}$

   (b) $\ddot{a}_{65|64}$

   (c) $\ddot{a}^{(12)}_{65|64}$ (Note: payment commences at the end of the month of (65)'s death, not the end of the year of (65)'s death.)

2. A life office sells an annuity to a man aged 65, for a single premium of £P. The payments are £10,000 per year, payable monthly in advance, and on his death a reduced widow's pension of £5,000 per year will be paid to his wife, now aged 61, provided she is then alive. The payments are guaranteed (at the initial level) for the first 5 years. Initial expenses are £100, and there is an administrative cost of £5 associated with each monthly payment. Both lives are subject to the mortality of the PMA92/PFA92 ultimate table, and the interest basis is 4% per annum. Calculate the single premium P.

3. (a) Express ${}_nq^2_{xy}$ in terms of single life probabilities and contingent probabilities referring to the first death.

   (b) Suppose that, for all lives, $\mu_x = (90 - x)^{-1}$. Evaluate ${}_{10}q^2_{30:40}$.

4. Define the following symbols in words, and give an expression for each of them in terms of an integral:

   (a) ${}_{\infty}q^1_{xy}$

   (b) $\bar{A}^2_{xy}$

   (c) $\bar{A}^1_{xy:\overline{n}|}$.

5. A husband and wife, aged 70 and 65 respectively, effect a policy under which the benefits are

   (a) a reversionary annuity of £10,000 p.a. payable annually in advance throughout the lifetime of the surviving spouse after the death of the first, and

   (b) a lump sum of £5,000 payable immediately on the second death.

   Level premiums are payable annually in advance until the first death. Calculate the annual premium on the following basis

   Males' mortality: PMA92 ultimate

   Females' mortality: PFA92 ultimate

   Interest: 4% p.a.

   Expenses: None.

   Ignore the possibility of divorce.